

Assignment 2 (same groups)

Objective

The main objective of this assignment is to produce a map of flooded areas for the Mondego River, from Coimbra to Figueira da Foz, using a combination of satellite image classifiers and citizen contributions. While in the first assignment the flooded area used was collected from the Copernicus Emergency Management Service; in this assignment, the objective is to produce a similar output by combining the results from satellite image classifiers and assess how contributions from citizens could potentially be incorporated.

Steps of the assignment

1. Download at least the Sentinel-2 tile (February 2026, cloud free) from the Copernicus Browser (<https://browser.dataspace.copernicus.eu/>) capturing the Mondego River in the region Coimbra- Figueira da Foz (the Sentinel-2 tile for this region is the TNE).
2. Train an image classifier with sentinel-2 data collected from benchmark data. Both with a “simple convolutional neural network” and following a transfer learning approach.
3. Apply the classifiers trained in step 2 to the Mondego and its vicinity from the Sentinel-2 image downloaded in step 1.
4. Discuss the results of step 3, including possible improvements, comparison with available data (if there is any), possible use in other epochs of imagers (different dates to monitor the flood), among other issues considered relevant.
5. Combine the data collected by citizens to the output of the image classification performed in step 3 and discuss the results.

Deliverables:

- Written report that should include:
 - 1) Training data preparation. Description of how the training data was selected and processed from the original benchmark datasets.
 - 2) Data pre-processing. Steps required to transform the original downloaded image (from step 2) into a form suitable for inference using the trained models;
 - 3) Model description. Description of the chosen model, including their characteristics and implementation details;
 - 4) Flood detection results. Presentation of the resulting raster with detected flooded areas and its quality metrics;
 - 5) Data fusion approach. Description of how citizen-contributed data was combined with the flooded raster map and motivating the used method;
 - 6) Performance comparison. comparison of quality metrics between the classification results alone and the results obtained after integrating citizen contributions.
- Files containing the flooded areas computed throughout the assignment.